

Monotone Classification with Relative Approximations

Lecture Notes for SC2320 Data Analytics and Mining

April 3, 2026

Instructions to readers

The flow of this set of notes is in tandem to the flow of the original paper. It will also go in the same flow as the presentation slides. Concepts/Material/Information that is auxiliary to the core ideas of this paper will be outlined at the end of each section.

Acronym Reference

Acronym	Full Form
ACR1	Acronym One
ACR2	Acronym Two
ACR3	Acronym Three

1 Background and Motivation

1.1 Entity Resolution

Entity Resolution is a process in Data Preprocessing and Integration that aims to address the issue of matching and merging records corresponding to the same real-world object.

Remark 1.1. Professor Tao's paper itself sometimes uses "Entity Matching" in the title and introduction to refer more broadly to the problem. For disambiguity in this set of notes, "**Entity Resolution**" will refer to the entire process while "Entity Matching" will refer to the process that the paper primarily focuses on.

1.2 The ER Pipeline

1.3 Matching in Depth

1.3.1 Why Human Labelling is Expensive

1.3.2 The Binary Classification Objective

1.3.3 The Monotonicity Rule

1.3.4 Active Classification / Learning

2 RPE Algorithm

2.1 The Algorithm

2.2 Why It Works

2.3 Guarantees and Limitations

3 Coresets (RCC)

3.1 Why RPE Isn't Enough

3.2 The Coreset Technique

3.3 Achieving the $1+\varepsilon$ Guarantee

4 Proofs and Lower Bounds

4.1 Lower Bound Results

4.2 Near-Optimality of the Algorithms

4.3 Mathematical Machinery

Summary

References

- [1] Y. Tao, “Monotone Classification,” *PODS’18*.
- [2] Y. Tao, “Monotone Classification (Extended),” *PODS’21*.